

Practical 1 - Review of linear regression and introduction to logistic regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Dataset required: `mental.RData`

Introduction

The purpose of this session is to perform a full analysis of the mental activity data trial introduced earlier. We will use a series of normal error models, and compare their respective fits using F-tests.

The data are from a three-arm placebo-controlled randomised trial assessing the effects of injection with morphine and heroin on mental activity. Seventy two people were randomized to one of three arms (placebo, morphine, heroin), and mental activity was assessed by the sum of scores on seven items of a questionnaire, with a high score indicating high mental activity.

The dataset contains three variables:

- `treat` - Randomised arm: 1=placebo; 2=morphine; 3=heroin
- `prement` - Mental activity score before injection
- `mentact` - Mental activity score 2 hours after injection with drug.

The questions that the analysis should address are as follows.

Aims

- Do the mean post-injection mental activity scores differ among groups?
- What effect does adjustment for pre-injection mental activity score have on your conclusions?

- Are the effects of the drugs different for subjects with different levels of pre-injection mental activity?

Analysis

Question 1

Read the data into R and examine. Use tables, histograms and plots to look at the distributions of the pre- and post-injection scores. Also plot the post-injection scores against the pre-injection scores, distinguishing between the three groups.

Discuss with your colleagues what conclusions can be drawn about the appropriateness of linear regression models for post-injection scores on the basis of these preliminary explorations.

Question 2

Write down in algebraic form the linear predictors that correspond to each of the models described in Table 1. Carefully define each term that you use in the various models.

Note that the distributional assumption in each model is $Y_i \sim N(\mu_i, \sigma^2)$ where Y_i is the post-injection mental activity score. In each model the link function is the identity $\eta_i = \mu_i$.

Table 1: Model linear predictors

#	Terms fitted	Linear predictor
1	Overall mean	
2	Drugs	
3	Pre-inj (Pre)	
4	Drugs + Pre	
5	Drugs + Pre + (Drugs-by-Pre interaction)	

Question 3

Fit each linear regression model using `lm` to complete Table 2.

Table 2: Residual sum of squares and degrees of freedom for fitted models

#	Terms fitted	RSS	Residual df
1	Overall mean		
2	Drugs		

#	Terms fitted	RSS	Residual df
3	Pre-inj (Pre)		
4	Drugs + Pre		
5	Drugs + Pre + (Drugs-by-Pre interaction)		

Question 4

Calculate the fitted values for model 5. Display these for each trial arm in a graph. Compute the intercepts and slopes of the three fitted lines from the parameters estimated in the model.

Table 3: Fitted intercepts and slopes by trial arm

Trial arm	Fitted intercept	Fitted slope
Placebo		
Morphine		
Heroin		

Question 5

Pen & paper exercise: use the results in Table 2 to compute test statistics to compare

- the fits of models 3 and 4 and
- the fits of models 4 and 5.

In each case convert the test statistic to a p-value using statistical tables (good practice for the summer examination!) or the `pnorm` function.

Confirm your results using the `anova` command.

Discuss: What do you conclude from these tests? Why can an analogous test not be used to compare the fits of models 2 and 3?

Question 6

Use results from models 2 and 4 to complete Table 4.

Table 4: Means and differences from placebo

Trial arm	Mean	Mean difference from placebo	SE	Adjusted difference from placebo	SE
Placebo		0	–	0	–
Morphine					
Heroin					

Question 7

Briefly summarise your conclusions regarding the effects of morphine and heroin on mental activity scores in this trial.

Question 8

If you have time, explore approaches for obtaining the estimate and confidence interval for the difference in mean post-treatment scores between the heroin and morphine groups in model 4. The coefficient estimates from a model can be extracted using `coef(mod4)`.